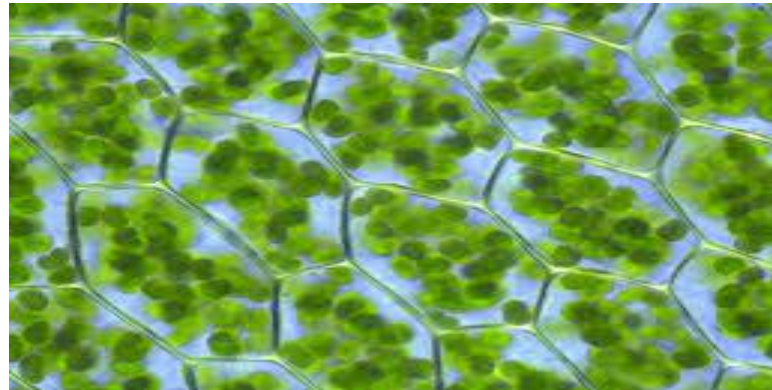
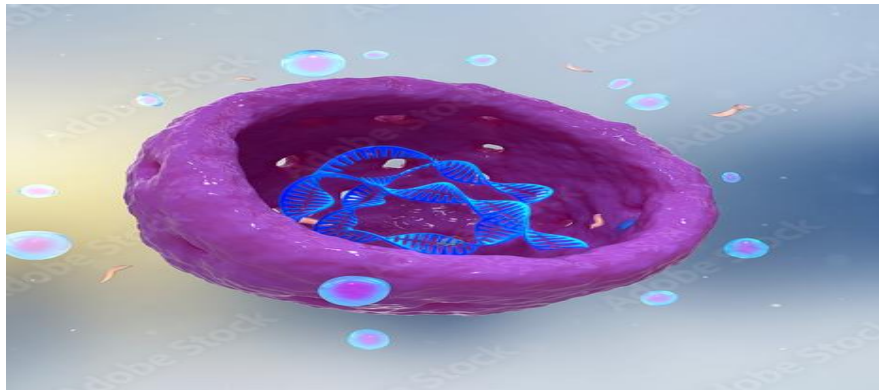




Cell Biology

Lecture 10



Dr. Sally Atef

Program/Course Learning Outcomes:

Program Outcomes (NARS 2017):

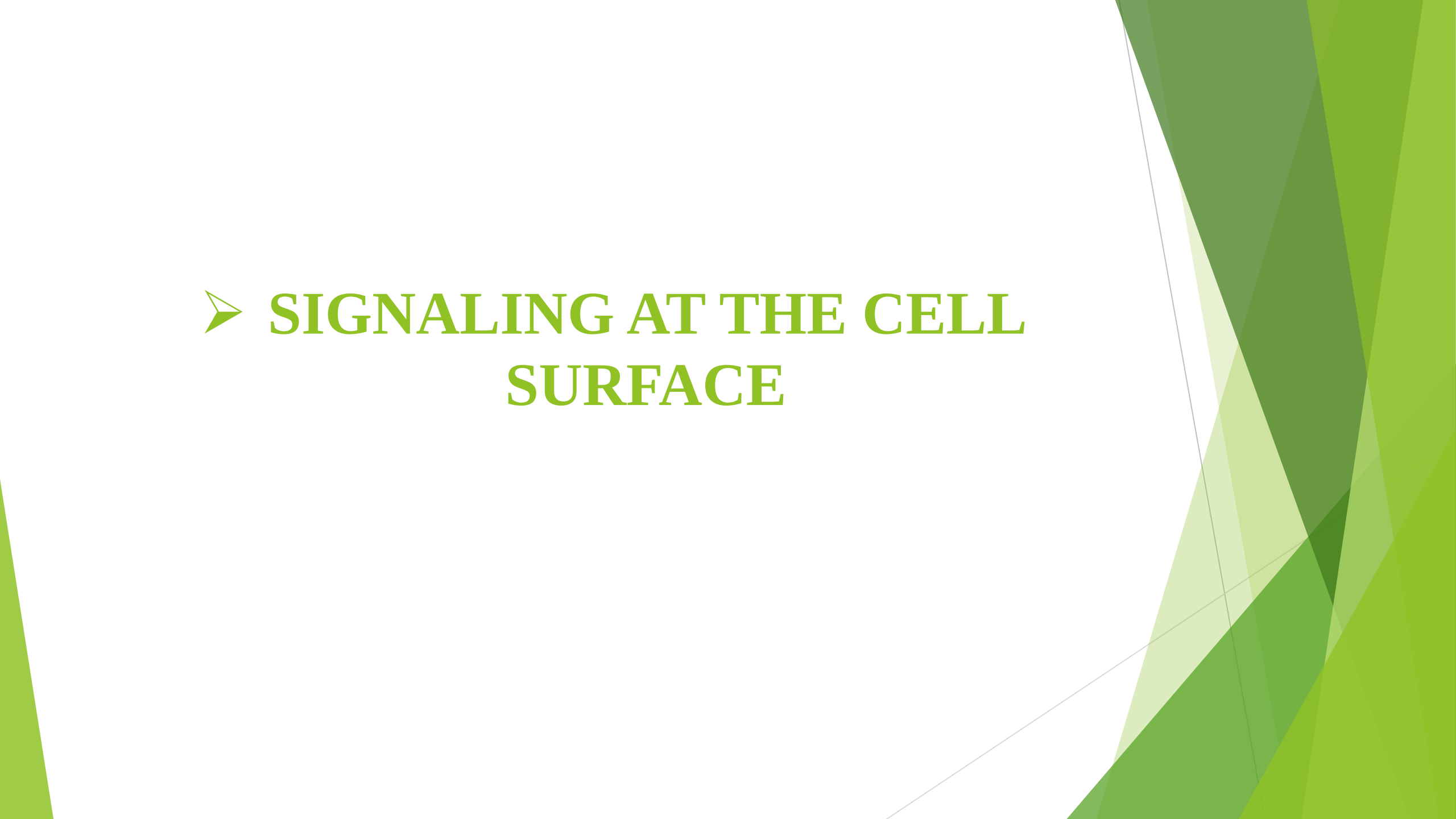
Demonstrate understanding of knowledge of pharmaceutical, biomedical, social, behavioral, administrative, and clinical sciences.

Course Learning Outcomes:

c1: Interpret principles of cell structures, cell signaling, gene regulation, protein synthesis and mutation in treatment of disease.

Lecture outline

- ▶ **SIGNALING AT THE CELL SURFACE**
- ▶ **Introduction**
- ▶ **Signaling Molecules and Cell-Surface Receptors**
- ▶ **Receptors**
- ▶ **Cell-Surface Receptors**

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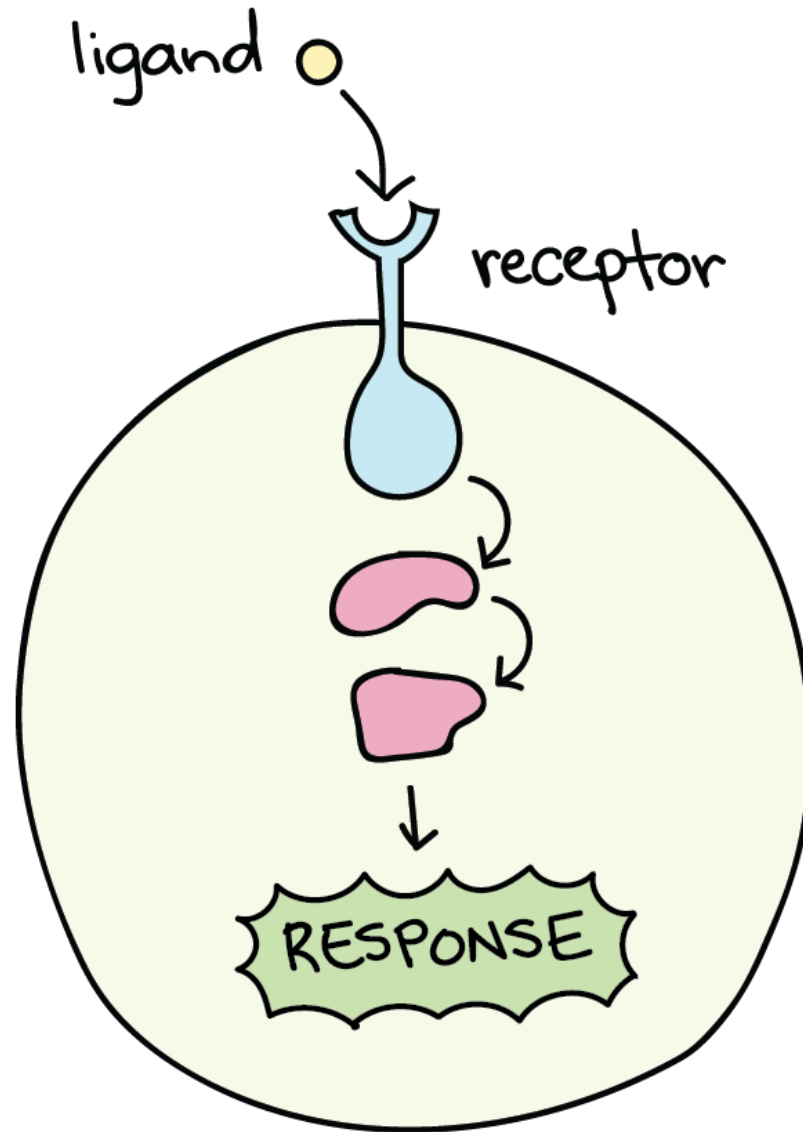
➤ SIGNALING AT THE CELL SURFACE

Introduction

- ▶ No cell lives in isolation.
- ▶ Extracellular signaling molecules are synthesized and released by signaling cells and produce a specific response only in target cells that have receptors for the signaling molecules.
- ▶ The overall process of converting signals into cellular responses, as well as the individual steps in this process, is termed **signal transduction**.

Introduction

- ▶ The signaling molecule acts as a **ligand**, which binds to a structurally complementary site on the extracellular or membrane-spanning domains of the receptor.
- ▶ Binding of a ligand to its receptor causes a conformational change in the cytosolic domain or domains of the receptor that ultimately induces **specific cellular responses**.



Chemical
messengers
relay the
signal

Types of Receptors

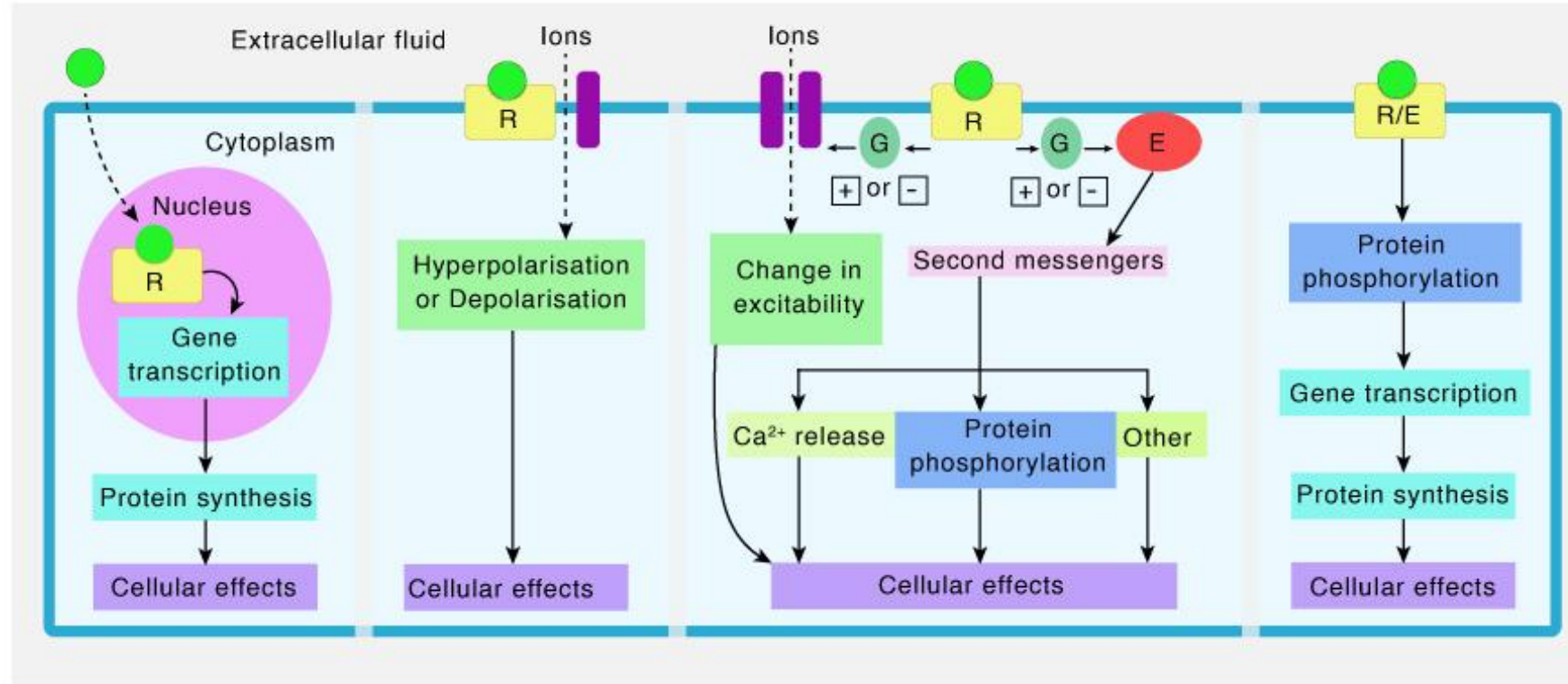
Intracellular receptor

Ligand-gated ion channel

Cell-surface receptor

G-protein coupled receptor

Enzyme-linked receptor

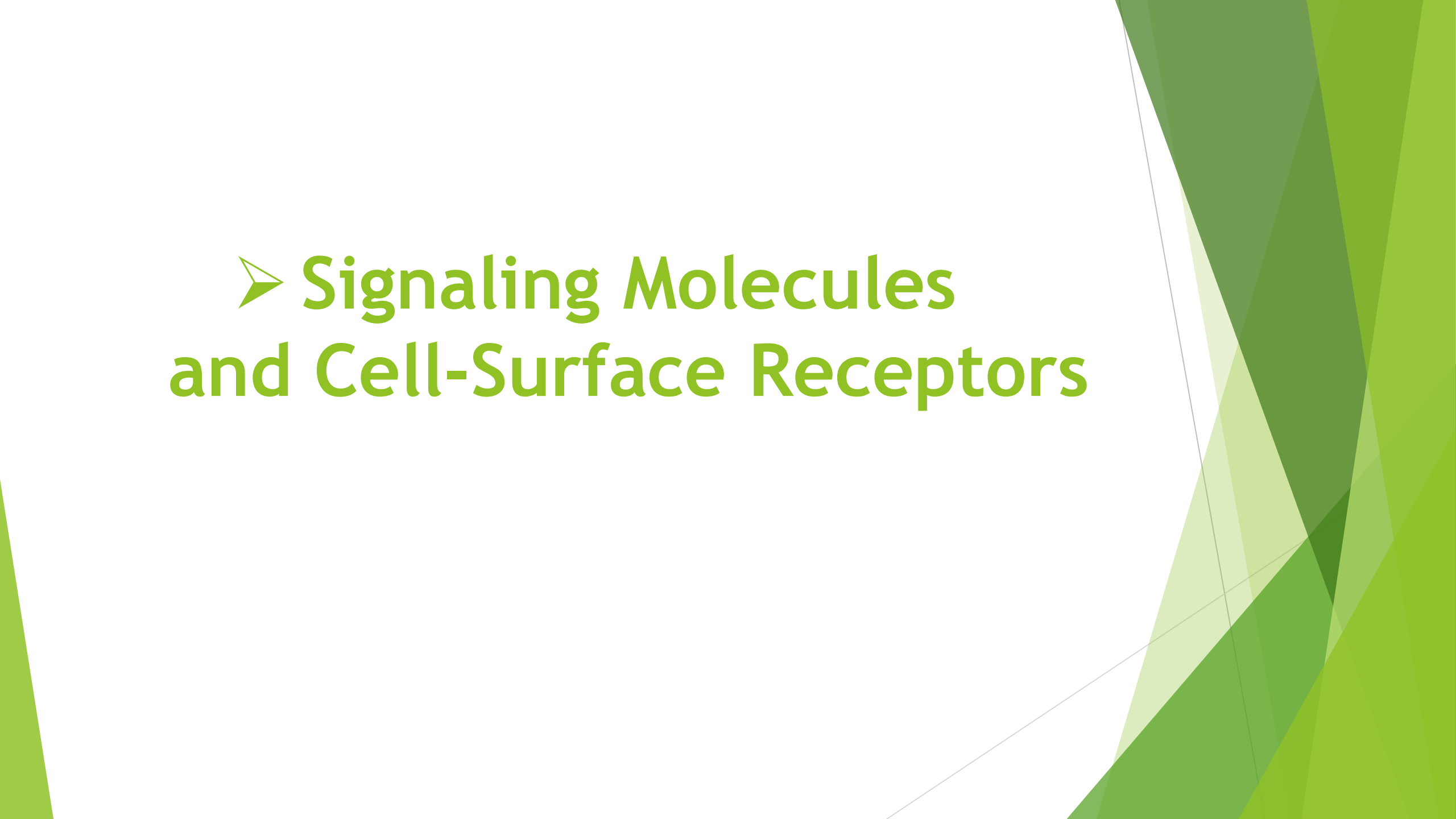


● → Signaling molecule

R → Receptor

G → G-protein

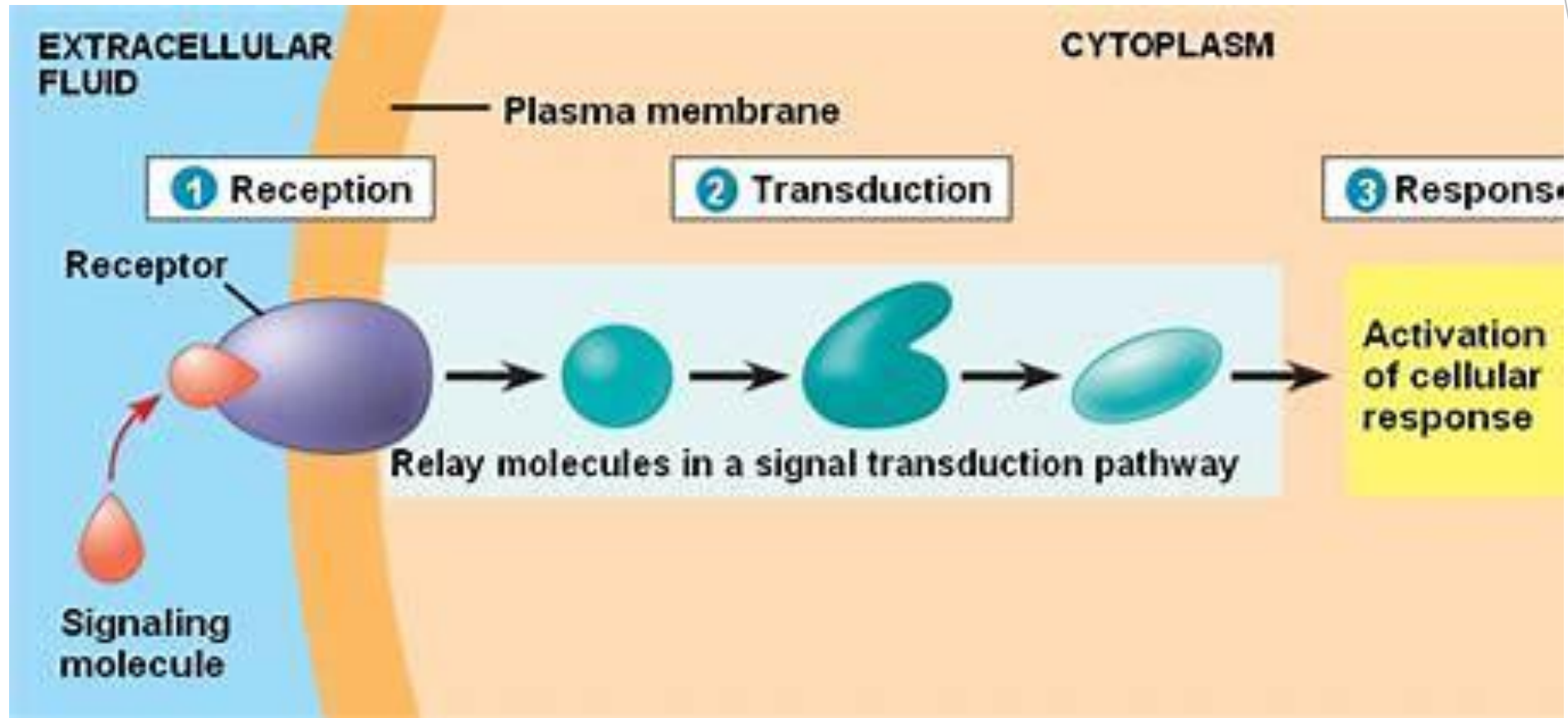
E → Effector

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➤ Signaling Molecules and Cell-Surface Receptors

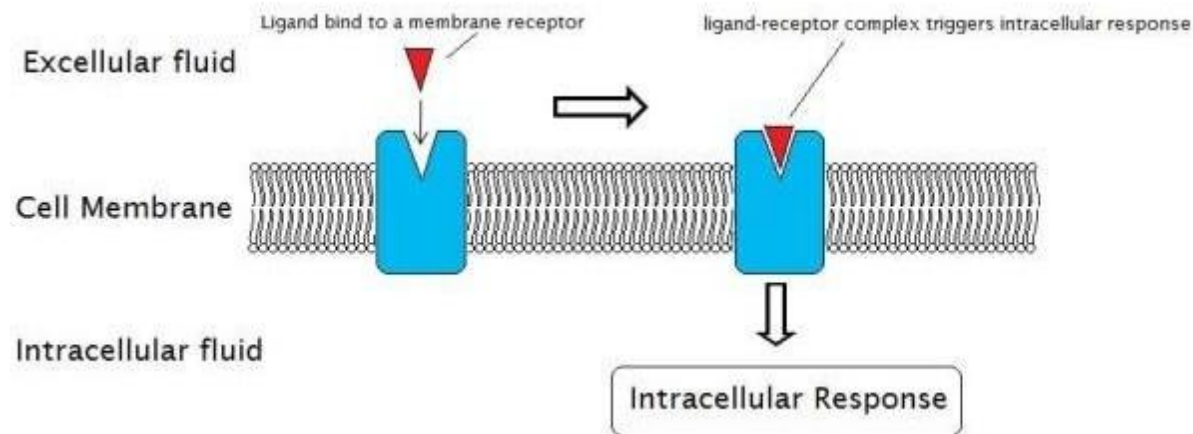
➤ Signaling Molecules and Cell-Surface Receptors

- ▶ **Communication by extracellular signals usually involves the following steps:**
- ▶ (1) Synthesis of the signaling molecule by the signaling cell
- ▶ (2) Release of the signaling molecule by the signaling cell
- ▶ (3) Transport of the signal to the target cell
- ▶ (4) Binding of the signal by a specific receptor protein leading to its activation
- ▶ (5) Initiation of one or more intracellular signal-transduction pathways by the activated receptor
- ▶ (6) Specific changes in cellular function, metabolism, or development
- ▶ (7) Removal of the signal, which often terminates the cellular response



➤ Receptors

- ▶ **A receptor** is a protein molecule that receives signals (**chemical or hormonal**) from outside the cell and is usually located on the cell surface.
- ▶ Such extracellular signals include hormones, cytokines, growth factors, neurotransmitters, lipophilic signaling molecules such as prostaglandins, and cell recognition molecules.



✓ Cell Surface Receptors

- ▶ **Cell surface receptors have the following components/domains:**
- ▶ 1. The **extracellular domain** which binds ligands and is exposed to the outer surface of the cell; also known as the **recognition domain**
- ▶ 2. The **membrane-spanning region** made up of hydrophobic protein molecules
- ▶ 3. The **intracellular domain** which is in contact with the cytoplasm and is involved in signal transduction; also known as the **coupling domain**

Cell surface receptors are generally classified into the following categories:

1. Ligand-gated ion channel-linked receptors
2. Enzyme-linked receptors
3. G-protein-linked receptors

